AGU23
San Francisco, CA & Online Everywhere
11-15 December 2023

[A51K-2075]

# Comparison of chemical reanalyses using daily ozone measurements by an ozone intrusion case

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## 1. Introduction

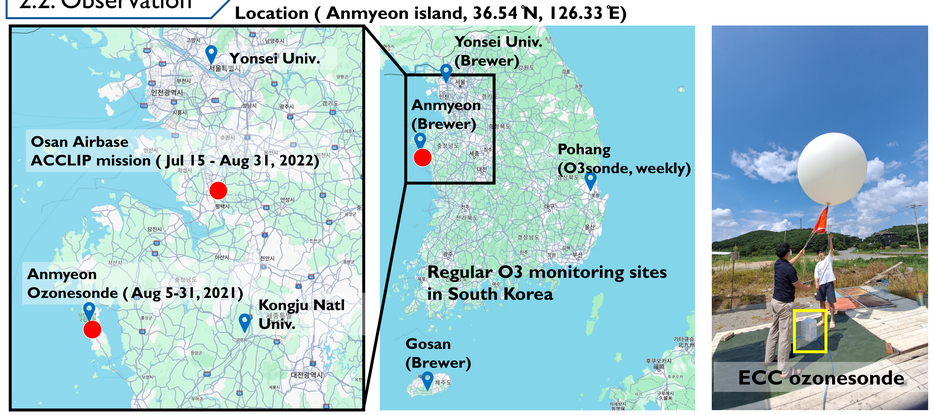
Stratosphere-Troposphere Exchanges (STE) is important for understanding chemical composition in the upper troposphere and lower stratosphere (UTLS; Holton et al. 1995). Ozone serves as a chemical tracer in the UTLS and can be used to detect STE events. Intrusion of ozone from the stratosphere, where ozone concentration is generally high, affects the troposphere ozone budget and related chemical processes. However, ozonesonde measurements are generally made on a weekly basis, it is too limited to capture synoptic scale STE events. In addition, recently provided chemical reanalyses require observation based evaluation. In this study, daily ozonesonde measurements were conducted, and two chemical reanalyses (MERRA-2 & CAMS) were evaluated using the ozonesonde measurements for a STE event measured in August 2021.

## 2. Data and Observation

### 2.1. Data

| Type | Data | Variable | Period | Resolution |
|---|---|---|---|---|
| Observation | ECC Ozonesonde | Ozone Partial Pressure, Ozone Mixing Ratio, Total Column Ozone | 2021.08.05 ~ 2021.08.31 | ~ 100 m |
| Atmospheric reanalysis | ERA5 | Geopotential, Potential Vorticity, U Velocity, V Velocity | | 1.5° × 1.5° |
| Chemical reanalysis | MERRA-2 | Ozone Mixing Ratio, Total Column Ozone, Pressure, U Velocity, V Velocity, Temperature | | 0.5° × 0.625° |
| | CAMS | | | 0.75° × 0.75° |

### 2.2. Observation




## 3. Result

### 3.1 Observation result & Ozone intrusion case

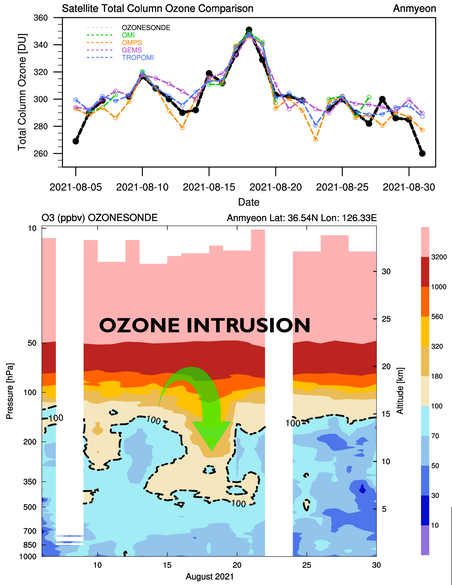



Figure 2. Time series of ozone concentration(shading, ppbv) from ozonesonde measurement. The black dash line is 100 ppbv ozone.

Figure 1. Time series of ozonesonde measurement total column ozone comparison with satellite observation total column ozone. Black dash line is Ozonesonde total column ozone (TCO), green dash line is OMI, orange dash line is OMPS, purple dash line is GEMS and blue dash line is TROPOMI.

During observation period, there is a decent ozone intrusion peak in Fig.1. STE event was associated with anticyclonic wave breaking. Due to the inflow by horizontal transport, high ozone concentration was observed in the middle layer of the troposphere.

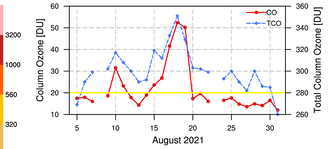



Figure 3. Time series of total column ozone and column ozone between 350 hPa and 100 hPa layer. Red solid line is column ozone. Blue dash line is total column ozone from ozonesonde measurement. Yellow solid line is standard of 20 DU.

Table 1. Classification of stratosphere troposphere exchange case.

| Condition | $Column\ Ozone_{100\sim350\ hPa} \geq 20\ DU$ |
|---|---|
| Case1 | 10 ~ 11 (2 days) |
| Case2 | 15 ~ 19 (5 days) |
| Background | The others (17 days) |

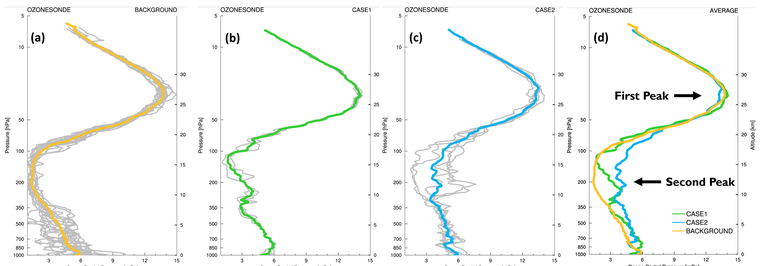



Figure 4. Vertical ozone concentration (mPa) structure by cases. Grey lines are respectively ozonesonde profiles. (a) is background, (b) is case1, (c) is case2, (d) is only average ozone concentration from each cases.

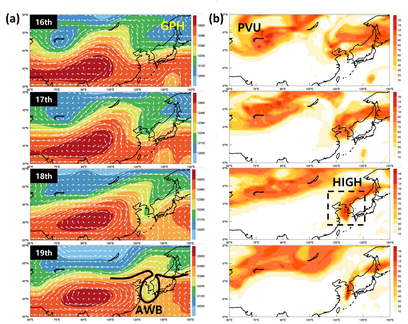



Figure 5. Composite maps of (a) 200-hPa geopotential height (m), (b) potential vorticity unit ($10^{-6}\ m^2\ s^{-1}\ kg^{-1}$) during period from 16th August to 19th August

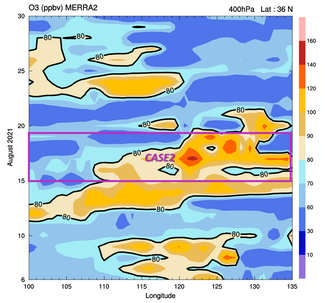



Figure 6. The Hovmöller diagram of the ozone concentration (shading, ppbv) at 400-hPa. Black solid line is 80 ppbv.

### 3.2. Comparison between Ozonesonde and Reanalyses (MERRA-2 & CAMS)

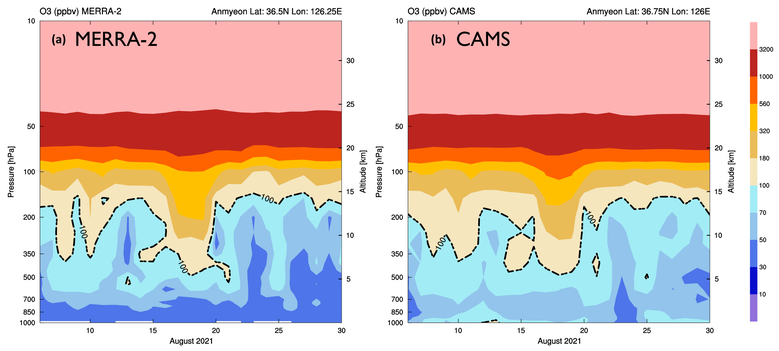



Figure 7. The same as Figure 2. but (a) is MERRA-2, (b) is CAMS

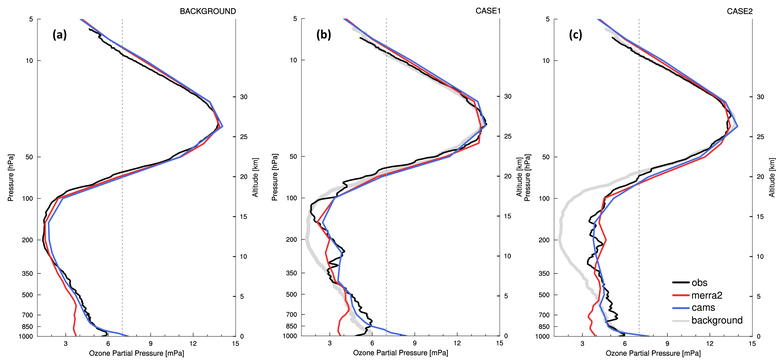



Figure 8. The same as Fig. 2. (a) is Background, (b) is case1, (c) is case2. Grey solid line is average ozone concentration of background.

Table 2. Statistical comparisons of the MERRA-2 and CAMS ozone against ozonesonde measurements by layers in observation period. The percent values are relative to ozonesonde value.

| Pressure | Value | Ozonesonde | MERRA-2 | CAMS |
|---|---|---|---|---|
| **Surface** 1000 hPa (mPa) | Mean | **5.66** | 3.78 (-33.2%) | 7.56 (33.7%) |
| | Standard Deviation | 1.54 | 0.18 (-88.6%) | 1.49 (-3.1%) |
| **Lower Troposphere** 850 hPa ~ 500 hPa (DU) | Mean | **18.30** | 15.46 (-15.5%) | 18.32 (0.1%) |
| | Standard Deviation | 4.76 | 2.82 (-40.8%) | 2.57 (-46.8%) |
| **UTLS** 350 hPa ~ 100 hPa (DU) | Mean | **21.41** | 24.91 (16.3%) | 30.34 (41.7%) |
| | Standard Deviation | 11.84 | 12.65 (6.9%) | 11.43 (-3.5%) |

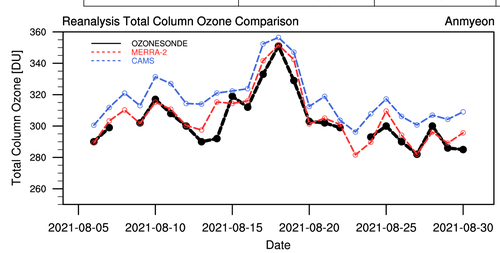



Figure 9. Time series of ozonesonde measurement total column ozone comparison with calculated chemical reanalysis total column ozone. Black dash line is ozonesonde total column ozone, red dash line is calculated MERRA-2 total column ozone and blue dash line is calculated CAMS total column ozone

Both of chemical reanalyses effectively represent a STE event. Comparing the ozone concentration vertically, CAMS value is closely following with the value from ozonesonde measurements. In terms of TCO, MERRA-2 closely matches the TCO value from ozonesonde measurements.

## 4. Summary

1. During the pre-Asian Summer Monsoon Chemical & Climate Impact Project (pre-ACCLIP) in 2021, a total of 26 vertical ozone measurements were conducted in Anmyeon, South Korea.
2. As STE event occurred, total column ozone significantly increased ~15%, reaching ~350 DU.
3. MERRA-2 underestimates the lower tropospheric ozone, and CAMS overestimates near surface ozone compared to observation.
4. For total column ozone, the value of MERRA-2 closely follows the ozonesonde value.

This research was supported by National Institute of Environmental Research (NIER2021-01-02-038) of Ministry of Environment in South Korea.